Online Fair Allocation of Indivisible Goods: Recent Progress and Open Questions

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Abstract

Fair division of indivisible goods has matured around relaxations of envy-freeness and proportionality, such as EF1, EFX, and MMS. Much less is settled when goods arrive online and must be allocated before future goods or valuations are known. This survey reviews models and algorithmic results for online fair allocation, with emphasis on indivisible goods. We organize the literature around four axes: arrival model, commitment model, information model, and fairness horizon. We then discuss online item-centric mechanisms, vanishing envy, reallocation for maintaining EF1, online MMS, stochastic and bandit-learning formulations, and recent best-of-many-worlds Nash-welfare guarantees. Throughout, we state representative results in theorem form and give complete proofs for foundational statements and proof sketches for technically deeper theorems whose full proofs are lengthy.

1 Introduction

Classical discrete fair division studies a fixed set of agents N and a fixed set of indivisible goods M, together with valuation functions $v_i : 2^M \to \mathbb{R}_{\geq 0}$. In the additive case, $v_i(S) = \sum_{g \in S} v_i(g)$. Exact envy-freeness and proportionality can fail even in the minimal instance with one good and two agents who both value it positively. This impossibility motivates relaxations such as EF1, EFX, and MMS, as reviewed by Amanatidis et al. [2].

Online fair allocation changes the nature of the problem. Goods may arrive sequentially, agents may arrive and depart, or both sides may be dynamic. Decisions may be irrevocable, batched, or reversible at some adjustment cost. Aleksandrov and Walsh [1] identified this online aspect as both a design opportunity and a source of impossibility: future uncertainty restricts what a mechanism can know, but it also suggests new online properties and new mechanisms such as LIKE, BALANCED LIKE, and PARETO LIKE.

The most common online indivisible-goods model is the following. There are n fixed agents. A sequence of indivisible goods $g_1, g_2, \ldots, g_T$ arrives over time. At time t, the algorithm observes some information about the arriving good, often the value vector

$$(v_1(g_t), \ldots, v_n(g_t)),$$

and must allocate g_t immediately to one agent, before seeing future goods. The final allocation after T rounds is $A^T = (A_1^T,$
$ldots, A_n^T)$, where A_i^T is the set of goods allocated to agent i. If the allocation is irrevocable, then once g_t is assigned, it cannot later be moved.

The offline AIJ 2023 survey naturally leads to the online agenda. In the offline world, EF1 is always attainable, MMS is approximately attainable, and EFX remains difficult. In the online world, the corresponding questions become sharper:

- Can fairness hold at the end, at every prefix, or only asymptotically?
- Can an irrevocable algorithm achieve EF1, MMS, or proportionality-type guarantees?

- How much reallocation is necessary to maintain fairness over time?
- How do stochastic arrivals, bandit feedback, or predictions change the picture?

2 Offline preliminaries

2.1 Allocations and valuations

A discrete fair allocation instance is (N, M, v), where $N = [n]$ is the set of agents, M is the set of goods, and $v = (v_1, \dots, v_n)$. An allocation is a tuple $A = (A_1, \dots, A_n)$ of pairwise-disjoint bundles whose union is M. Unless explicitly stated otherwise, valuations are nonnegative, normalized, monotone, and additive.

Definition 2.1 (Envy-freeness)**.** *An allocation A is envy-free (EF) if*

$$v_i(A_i) \geq v_i(A_j) \qquad \forall i, j \in N.$$

Definition 2.2 (EF1)**.** *An allocation A is envy-free up to one good (EF1) if for every pair $i, j \in N$, there exists a good $g \in A_j$ such that*

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}).$$

If $A_j = \emptyset$, the condition is interpreted as true.

Definition 2.3 (EFX)**.** *An allocation A is envy-free up to any good (EFX) if for every pair $i, j \in N$ and every good $g \in A_j$,*

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}).$$

Definition 2.4 (Maximin share)**.** *For agent i, the maximin share value is*

$$\mu_i^n(M) = \max_{(B_1,\dots,B_n)} \min_{k \in [n]} v_i(B_k),$$

where the maximum is over all partitions of M into n bundles. An allocation A is MMS if

$$v_i(A_i) \geq \mu_i^n(M) \qquad \forall i \in N.$$

For $\alpha \in [0, 1]$, A is α-MMS if $v_i(A_i) \geq \alpha \mu_i^n(M)$ for every i.

2.2 A baseline: round-robin implies EF1

The simplest offline algorithm repeatedly lets agents pick their favorite remaining good in a fixed cyclic order.

Theorem 2.5 (Round-robin EF1, Caragiannis et al. [5])**.** *For additive valuations, round-robin returns an EF1 allocation.*

Proof. Fix two agents i and j. Suppose first that i precedes j in the cyclic order. In each round, when i chooses, all goods later chosen by j in that round are still available. Hence the good chosen by i in a given round is worth at least as much to i as the good subsequently chosen by j in the same round. Summing over rounds gives $v_i(A_i) \geq v_i(A_j)$, so i does not envy j.

If j precedes i, let g be the first good chosen by j. After removing g from A_j, every remaining good in A_j can be matched to a later-round good chosen by i before j's corresponding pick. Thus $v_i(A_i) \geq v_i(A_j \setminus \{g\})$. Since this holds for every ordered pair, the allocation is EF1. □

This proof is often the first point at which the online difficulty becomes visible: round-robin assumes each agent can choose among all remaining goods. In an online setting, the future choice set does not yet exist.

3 Online models and taxonomy

The online fair allocation literature can be organized along four mostly independent dimensions.

3.1 Arrival model

- **Adversarial arrivals:** the sequence of goods and value vectors may be worst-case. Competitive analysis and impossibility results dominate this model.
- **Stochastic arrivals:** goods, types, or values are sampled from distributions. One may seek high-probability or expected guarantees.
- **Nonstationary stochastic arrivals:** distributions may change over time, but not adversarially without restriction.
- **Learned or predicted inputs:** the mechanism is given predictions about future goods, optimal outcomes, or distributional parameters.

3.2 Commitment model

Definition 3.1 (Irrevocable online allocation)**.** *An online algorithm is irrevocable if the assignment of g_t at time t cannot be changed at any later time.*

Some models allow reallocations. If A^t and A^{t-1} denote consecutive allocations, the number of adjustments at time t is commonly measured by the number of old goods whose owner changes between the two allocations. This gives a bridge between fully offline recomputation and fully irrevocable online allocation.

3.3 Information model

At the arrival of g_t, the algorithm may know the full value vector, only ordinal rankings, binary likes/dislikes, stochastic estimates, or bandit feedback. The strongest full-information online model is still difficult; weaker information models introduce learning regret and constraint-violation tradeoffs.

3.4 Fairness horizon

- **Final-time fairness:** the allocation after T rounds should be fair.
- **Prefix-wise fairness:** the allocation after every prefix $t \leq T$ should be fair.
- **Asymptotic fairness:** normalized unfairness, e.g. Envy_T/T, should vanish as $T \to \infty$.
- **Ex ante fairness:** fairness is imposed on expected utilities or random shares.
- **Ex post fairness:** fairness must hold for every realized allocation.

4 Online mechanisms for arriving indivisible goods

The AAAI 2020 survey emphasized that many offline mechanisms are agent-centric: agents choose from a set of currently available goods. Online indivisible allocation suggests an item-centric view: when a good arrives, decide which agent is eligible to receive it, then possibly randomize among eligible agents.

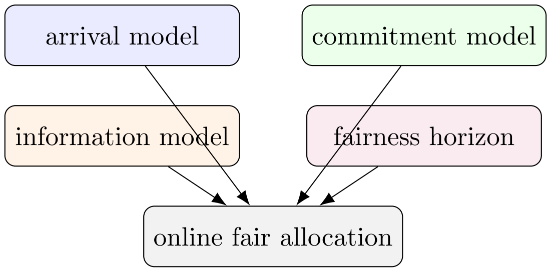

Figure 1: Four axes that help classify online fair allocation models.

4.1 LIKE-type mechanisms

Assume agents report a bid or utility b_{it} for arriving item g_t. In binary instances, $b_{it} \in \{0, 1\}$.

Definition 4.1 (LIKE)**.** *The LIKE mechanism allocates g_t uniformly at random among agents with $b_{it} > 0$. If no agent has positive bid, the item may be discarded or allocated by a tie-breaking rule.*

Definition 4.2 (BALANCED LIKE)**.** *BALANCED LIKE allocates g_t uniformly at random among agents with $b_{it} > 0$ who currently have the smallest number of allocated items among the agents with positive bid.*

LIKE is simple and online strategy-proof in binary-like settings: an agent who likes an item cannot reduce current expected utility by declaring a positive bid. BALANCED LIKE improves balance of item counts, but the balancing rule can create strategic incentives to skip a current item in the hope of becoming more eligible later.

Proposition 4.3 (Online strategy-proofness of LIKE in binary instances)**.** *In binary utility instances, if an arriving item is allocated uniformly among agents declaring value 1, then truthful reporting is a weakly dominant one-step action for the current round.*

Proof. Fix a round t, an agent i, and the other agents' declarations. If $v_i(g_t) = 0$, agent i's utility from receiving the item is zero, so either report gives expected utility zero for this item. If $v_i(g_t) = 1$, reporting 0 gives probability zero of receiving the item, while reporting 1 gives probability $1/k$ for some $k \geq 1$, hence expected utility $1/k > 0$. Thus reporting truthfully is weakly optimal in the current round. The statement is one-step: it ignores future strategic effects, exactly as in the online strategy-proofness relaxation discussed by Aleksandrov and Walsh [1]. $\square$

5 Irrevocable allocation and vanishing envy

A central insight of Benade, Kazachkov, Procaccia, and Psomas [4] is that exact envy-type guarantees are too strong under irrevocable online allocation, but normalized envy can vanish over time.

Definition 5.1 (Cumulative envy)**.** *For an allocation after T rounds, define the envy of agent i toward agent j by*

$$E_{ij}^T = v_i(A_j^T) - v_i(A_i^T).$$

The maximum positive envy is

$$\mathrm{Envy}_T = \max_{i,j \in N} \max\{0, E_{ij}^T\}.$$

An algorithm has vanishing envy if $\mathrm{Envy}_T/T \to 0$ *as* $T \to \infty$.

Theorem 5.2 (Random allocation has vanishing envy)**.** *Suppose $v_i(g_t) \in [0,1]$ for all i, t. Allocate every arriving item independently and uniformly at random among the n agents. Then for every $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\mathrm{Envy}_T \le \sqrt{2T \log \frac{n(n-1)}{\delta}}.$$

Consequently, $\mathrm{Envy}_T/T \to 0$ *in probability.*

Proof. Fix an ordered pair (i,j). For each item g_t, define

$$X_t = v_i(g_t)\left(\mathbf{1}\{g_t \text{ is allocated to } j\} - \mathbf{1}\{g_t \text{ is allocated to } i\}\right).$$

Then $E_{ij}^T = \sum_{t=1}^T X_t$. Since each item is allocated uniformly and independently, $\mathbb{E}[X_t] = 0$, and since $v_i(g_t) \in [0,1]$, we have $X_t \in [-1,1]$. Hoeffding's inequality gives

$$\Pr\left[E_{ij}^T \ge r\right] \le \exp\left(-\frac{r^2}{2T}\right).$$

Taking a union bound over all $n(n-1)$ ordered pairs, with probability at least $1-\delta$ no pair has envy exceeding

$$r = \sqrt{2T \log \frac{n(n-1)}{\delta}}.$$

This proves the stated bound. □

The theorem is intentionally elementary. Benade et al. [4] prove stronger and essentially optimal bounds: they give a deterministic polynomial-time algorithm with $\mathrm{Envy}_T \in \widetilde{O}(\sqrt{T/n})$ and matching lower bounds up to logarithmic factors, including batch-arrival variants.

Theorem 5.3 (Benade et al. [4], informal)**.** *In the adversarial online allocation of T indivisible goods with values in* $[0,1]$*, there is a deterministic polynomial-time algorithm with*

$$\mathrm{Envy}_T \in \widetilde{O}(\sqrt{T/n}),$$

and this dependence is asymptotically optimal up to logarithmic factors against sufficiently powerful adversaries.

Proof idea. The positive result balances envy as a discrepancy-minimization process. Each allocation decision affects pairwise envy vectors, and the algorithm chooses recipients so that an appropriate potential function measuring cumulative pairwise imbalance grows slowly. The lower bound constructs adversarial sequences whose value vectors force any online algorithm to accumulate discrepancy of order roughly $\sqrt{T/n}$. The full proof uses martingale/discrepancy arguments and is substantially more technical than the elementary random-allocation bound above. □

6 Maintaining EF1 by changing the past

Irrevocable allocation is attractive operationally, but prefix-wise EF1 can be too demanding. He, Procaccia, Psomas, and Zeng [7] study the model where each prefix allocation must be EF1, but old goods may be reallocated. The goal is to minimize the number of adjustments.

Definition 6.1 (Adjustment)**.** *When item g_t arrives, the algorithm may transform the previous allocation of $g_1, \ldots, g_{t-1}$ into a new allocation of $g_1, \ldots, g_t$. Each previously allocated item whose owner changes counts as one adjustment.*

Theorem 6.2 (He et al. [7], EF1 with adjustments)**.** *For online allocation of T indivisible goods with additive valuations, where EF1 is required after every round:*

(i) for two agents, informed algorithms can maintain EF1 with no adjustments, while uninformed algorithms require $\Theta(T)$ adjustments in the worst case;

(ii) for three or more agents, even informed algorithms require $\Omega(T)$ adjustments, and there is an uninformed algorithm using $O(T^{3/2})$ adjustments.

Proof idea. For two agents, future information lets the algorithm commit to a final EF1-compatible ordering of items and reveal the corresponding prefix allocation without moving old goods. Without future information, an adversary can repeatedly present items that make the previous balance wrong, forcing linear many repairs. For $n \geq 3$, the need to keep all pairwise envy relations EF1-compatible creates cycles of obligations that cannot be anticipated even with full future knowledge, giving an $\Omega(T)$ lower bound. The $O(T^{3/2})$ upper bound is obtained by batching and periodically recomputing fair allocations, trading local unfairness control against adjustment cost. $\square$

This result clarifies a useful methodological point. The online problem is not simply "harder EF1". It introduces a new resource: stability of past allocations. The algorithmic goal becomes bicriteria: maintain fairness while minimizing disruption.

7 Online MMS and competitive guarantees

MMS is inherently global: $\mu_i^n(M)$ depends on all goods. In an online setting, the benchmark is usually the offline MMS value of the final instance, while the algorithm must allocate goods without seeing the future.

Definition 7.1 (Competitive online MMS for goods)**.** *An online algorithm is α-competitive for MMS goods if for every arrival sequence and every agent i, the final bundle satisfies*

$$v_i(A_i^T) \geq \alpha\, \mu_i^n(M_T),$$

where $M_T = \{g_1, \ldots, g_T\}$ is the full set of arrived goods.

Zhou, Bai, and Wu [12] connect this problem with online scheduling. For identical valuations, online MMS for goods corresponds to semi-online machine covering; for chores, it corresponds to online load balancing. General additive valuations are much more difficult.

Theorem 7.2 (Zhou, Bai, and Wu [12])**.** *For online allocation of indivisible goods with additive normalized valuations:*

(i) no positive competitive ratio is possible for MMS when $n \geq 3$, even with three agents;

(ii) for two agents, an optimal $1/2$-competitive online algorithm exists.

For chores, they give a $(2 - 1/n)$-competitive algorithm for $n \geq 3$, a $\sqrt{2}$-competitive algorithm for two agents, and a lower bound of $15/11$ for two agents.

Proof idea. The negative result for goods uses adversarial continuation. The algorithm's early irrevocable allocations reveal which agent can be made deficient. Future goods are then chosen so that the offline MMS benchmark for that agent becomes positive while the algorithm cannot retroactively give her enough value. The two-agent positive result exploits the special structure of two-way partitions: assigning each incoming good according to a balancing rule can ensure that the final utility of each agent is at least half of her offline maximin share. The matching lower bound constructs sequences where any online rule must choose between two symmetric futures, one of which makes the chosen allocation lose a factor of two. $\square$

The theorem shows a sharp distinction between EF1-type and MMS-type requirements. EF1 is pairwise and local; MMS is global and benchmark-dependent. Online irrevocability is especially damaging to global share guarantees.

8 Binary and bivalued valuations

Restricted valuation domains can restore positive results. Wang and Wei [9] study online allocation of goods and chores under binary and personalized bivalued valuations, with EF1, MMS, and utilitarian social welfare as objectives.

Definition 8.1 (Binary and personalized bivalued valuations)**.** *Binary valuations satisfy* $v_i(g) \in \{0, 1\}$*. Personalized bivalued valuations allow each agent* i *to have two possible values, often written* $\{a_i, b_i\}$*, for goods or chores, with values depending on both the agent and the item.*

Theorem 8.2 (Wang and Wei [9], summary)**.** *For online allocation of indivisible goods or chores under binary and personalized bivalued valuations, several online fairness-efficiency combinations that are impossible in unrestricted domains become achievable, while matching counterexamples show that the obtained guarantees are close to best possible.*

Proof idea. Binary values make each item relevant only to the agents who like it. This permits item-centric allocation rules that balance the number or accumulated value of liked items among eligible agents. Bivalued domains retain enough order structure to extend some balancing arguments, but counterexamples arise when personalized values make locally balanced decisions globally inefficient or unfair. □

This line is close in spirit to classical LIKE/BALANCED-LIKE mechanisms, but it evaluates stronger discrete-fairness notions rather than only ex ante envy or item-count balance.

9 Stochastic, bandit, and best-of-many-worlds models

9.1 Bandit learning with proportionality constraints

Schiffer and Zhang [8] study a model with finitely many item types, random arrivals, and initially unknown mean values. The algorithm must learn values while allocating indivisible items and satisfying proportionality in expectation.

Definition 9.1 (Proportionality in expectation)**.** *Let* U_i^T *be agent* i*'s cumulative utility after* T *rounds. A randomized online algorithm is proportional in expectation if*

$$\mathbb{E}[U_i^T] \geq \frac{1}{n}\mathbb{E}\left[\sum_{t=1}^{T} v_i(g_t)\right]$$

for each agent i*, up to the precise normalization and feasibility constraints of the model.*

Theorem 9.2 (Schiffer and Zhang [8])**.** *In the finite-type stochastic online fair division model with unknown mean valuations, there is a UCB-style algorithm that satisfies proportionality constraints with high probability and achieves* $\widetilde{O}(\sqrt{T})$ *regret for social welfare.*

Proof idea. The algorithm maintains upper confidence bounds for unknown mean values, then solves linear optimization problems that encode both welfare maximization and proportionality feasibility. Confidence intervals control estimation error uniformly over time, while the structure of proportionality constraints prevents the number of near-tight constraints from causing worse than $\widetilde{O}(\sqrt{T})$ regret. This improves the previous $\widetilde{O}(T^{2/3})$ rate. □

9.2 PACE and best-of-many-worlds guarantees

A different strand evaluates long-run fairness and efficiency through Nash welfare and multiplicative envy. Yang, Liao, and Kroer [10] analyze PACE, an integral greedy algorithm related to pacing dynamics.

Definition 9.3 (Multiplicative envy). *For positive utilities, the multiplicative envy of i toward j can be measured by a ratio such as*

$$\frac{v_i(A_j)}{v_i(A_i)},$$

with suitable smoothing or normalization when the denominator is zero.

Definition 9.4 (Nash welfare). *For a final allocation A, the Nash welfare is*

$$\mathrm{NSW}(A) = \left(\prod_{i=1}^{n} v_i(A_i)\right)^{1/n},$$

with standard conventions for zero utilities or positive-support variants.

Theorem 9.5 (Yang, Liao, and Kroer [10], summary). *For sequentially arriving items with adversarially chosen values subject to natural restrictions, integral greedy allocation/PACE achieves asymptotically bounded multiplicative envy and competitive Nash-welfare guarantees. These results complement earlier guarantees for stationary and stochastic-but-nonstationary arrivals, yielding a best-of-many-worlds picture.*

Proof idea. PACE allocates items using inverse-utility-like multipliers: agents with smaller accumulated utility receive larger pacing weights. The analysis views the dynamics as an online primal-dual or mirror-descent-like process. Potential functions bound the deviation between realized integral allocations and the target fair allocation trajectory, yielding simultaneous fairness and welfare guarantees across several input models. □

10 Online divisible allocation and related models

Although this survey focuses on indivisible goods, online divisible allocation remains an important comparator. Gkatzelis, Psomas, and Tan [6] study immediately and irrevocably allocated divisible resources under normalized valuations. They show that nontrivial guarantees are impossible in fully adversarial settings without normalization, while normalization permits strong welfare guarantees subject to fairness constraints for two agents.

Learning-augmented mechanism design is another related direction. The uploaded NeurIPS 2025 paper by Aziz et al. [3] concerns facility location rather than allocation of arriving indivisible goods. Still, it is methodologically relevant: it studies consistency–robustness tradeoffs under predictions, proving tight deterministic bounds and improved randomized mechanisms for an envy-ratio fairness objective.

11 Summary table

12 Open problems

Open Problem 12.1 (Online EF1 without reallocation). *Characterize the valuation domains and arrival models under which an irrevocable online algorithm can guarantee final-time EF1, prefix-wise EF1, or approximate EF1.*

Direction	Representative papers	Main message
Online survey and taxonomy	Aleksandrov–Walsh [1]	Online fair division varies by online agents/resources, resource divisibility, information, and mechanism commitment.
Vanishing envy	Benade et al. [4]	Irrevocable allocation can make exact envy control impossible, but envy per round can vanish.
Reallocation for EF1	He et al. [7]	Maintaining EF1 at every prefix is possible with reallocations; adjustment complexity is the key cost.
Online MMS	Zhou–Bai–Wu [12]	MMS is fragile online: no positive competitive ratio for goods with $n \geq 3$, but sharp results exist for two agents and chores.
Binary/bivalued domains	Wang–Wei [9]	Restricted valuation domains enable stronger online EF1/MMS/welfare combinations.
Bandit learning	Schiffer–Zhang [8]	Unknown valuation means can be learned with $\widetilde{O}(\sqrt{T})$ regret while maintaining proportionality constraints.
PACE and Nash welfare	Yang–Liao–Kroer [10]	Pacing-style greedy algorithms provide robust long-run fairness and Nash-welfare guarantees.

Table 1: Representative post-2020 directions in online fair allocation, especially for indivisible goods.

Open Problem 12.2 (Online MMS beyond impossibility). *For general additive goods, no positive MMS competitive ratio exists for $n \geq 3$. What relaxations make the problem meaningful: stochastic arrivals, resource augmentation, limited reallocation, predictions, or restricted valuations?*

Open Problem 12.3 (EFX online). *The offline existence of EFX for additive goods is already open for $n \geq 4$. Are there useful online analogues of approximate EFX that admit nontrivial asymptotic or stochastic guarantees?*

Open Problem 12.4 (Best-of-both-worlds online fairness). *Can one design algorithms that simultaneously provide ex post EF1-like guarantees for realized allocations and ex ante proportionality or envy-freeness guarantees in expectation?*

Open Problem 12.5 (Learning-augmented online fair allocation). *Develop consistency–robustness theory for online indivisible-goods allocation using predictions of future item types, values, or target fair shares.*

13 Concluding remarks

Online fair allocation of indivisible goods is not a minor variant of offline fair division. It changes both feasibility and the meaning of fairness. The offline theory asks whether a fair allocation exists and how to compute it. The online theory asks when fairness can survive uncertainty, irrevocability, and learning. The emerging picture is fragmented but coherent: local fairness notions such as EF1 interact naturally with reallocation; global benchmarks such as MMS are brittle under adversarial irrevocable arrivals; stochastic and bandit models recover useful guarantees; and pacing-style algorithms provide promising long-run fairness-welfare tradeoffs.

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